A Calibrated Separation of the Kontorovich–Lagarias and Volkov Exponent Predictions for the $5x + 1$ Reverse Tree

Renato Augusto Tavares
Universidade Federal de Goiás
https://orcid.org/0009-0002-0196-3311

August 12, 2026

Abstract

Kontorovich and Lagarias predict a counting exponent $\eta \approx 0.650919$ for the $5x + 1$ reverse tree under their un-accelerated map; the accelerated count $N_u(x) = x^{\eta+o(1)}$ used here shares it. A competing model due to Volkov predicts $\eta^* \approx 0.678$. Kontorovich and Lagarias write that Volkov's own data seems insufficient to discriminate between the values.

We enumerate the reverse tree exactly and extrapolate the truncation bias with Aitken's Δ^2. The fixed-window reading is 0.639; it excludes neither prediction. Applied to an unconstrained construction built to a known pressure-equation root, the same estimator under-reads by 0.038, more than the gap $\Delta = 0.027$ between the two disputed values, so a raw reading settles nothing.

Two more constructions share the arithmetic tree's offspring law and, via an exact sibling congruence, its sibling spacing, drawing the branch residue instead of the true one: an integer recursion and its real-valued relaxation, both built to exponent 0.650919. Together with the unconstrained construction above, three constructions built to 0.650919 read within 0.0035 of that value, a band 0.0037 wide. The arithmetic tree reads 0.64926, below the two constructions built to 0.650919 that share the sibling congruence, and inside the wider band set by the third. A fourth construction, the same real-valued relaxation built to exponent 0.678, reads 0.0282 away, more than seven widths of that band outside it.

The comparison is a measurement, not a proof, calibrated against constructions built to a known pressure-equation root. It targets the disputed exponent value; whether it bears on Volkov's branching model itself is a separate question.

1 Introduction

Kontorovich and Lagarias [1] study the $3x + 1$ and $5x + 1$ Collatz-type problems through the counting statistics of the reverse tree: for a fixed root u, their integer count $\pi_{u,5}(x)$, under their un-accelerated map (§2's Remark), conjecturally satisfies $\pi_{u,5}(x) = x^{\eta+o(1)}$ for an exponent η independent of u, their Conjecture 7.2; the accelerated count $N_u(x)$ used here shares that exponent. Their stochastic model predicts $\eta_{5,BP} \approx 0.650919$ for $q = 5$ (their Theorem 8.10, subscript BP for branching process, in their own notation). A competing branching model due to Volkov [3], discussed in the same reference, grows a complete binary tree under a different encoding of the iterates and predicts $\eta^*_{5,BP} \approx 0.678$, a gap of $\Delta = 0.027$. Volkov's model counts a related but different quantity, $Q(x)$: non-divergent trajectories entering some cycle. Kontorovich and Lagarias note that their own integer count is bounded by $Q(x)$, with the two orders of growth only conjectured to agree. Kontorovich and Lagarias write that "the empirical data Volkov presents seems insufficient to discriminate between these two predicted exponents," and separately note "some controversy concerning the conjectured value of the constant."

We measure the exponent by exact enumeration of the accelerated reverse tree. The fixed-window reading is 0.639, excluding neither prediction. Calibrated against constructions built to a known pressure-equation root, the arithmetic tree's reading falls inside the band set by three constructions built to 0.650919, and 0.0282 from a construction built to 0.678, more than seven band-widths away.

2 Setup: the reverse tree of the accelerated $qx + 1$ map

Fix an odd integer $q \geq 3$. The *accelerated $qx + 1$ map* is

$$T_q(n) = \frac{qn+1}{2^{v_2(qn+1)}}, \qquad n \text{ odd},$$

where v_2 is the 2-adic valuation. The *reverse tree* rooted at an odd integer u coprime to q is built by, at each odd node v, adjoining a child $w_a := (2^a v - 1)/q$ for every $a \geq 1$ such that $2^a v \equiv 1 \pmod q$; such w_a is automatically an odd integer whenever it is an integer. Writing $d := \operatorname{ord}_q(2)$, admissibility of a for a given parent v depends only on $v \bmod q$: if some $a_0(v) \in \{1, \ldots, d\}$ satisfies $2^{a_0(v)} v \equiv 1 \pmod q$, it is unique, and the admissible exponents are exactly $a \equiv a_0(v) \pmod d$; nodes whose residue admits no such a_0 are sterile. At $q = 5$, $d = 4$, every nonzero residue admits an a_0, and only $v \equiv 0 \pmod 5$ is sterile.

Every node has exactly one parent under T_q, so a given integer reaches u, if at all, along a single path. We count them:

$$N_u(x) := \#\{n \leq x : n \text{ odd and } T_q^k(n) = u \text{ for some } k \geq 0\}, \qquad N_u(x) = x^{\eta + o(1)}.$$

Remark 2.1. Kontorovich and Lagarias's Theorem 8.10 is a statement about the branching random walk $B[5^0]$: almost surely, its progeny count $I^*(x;\omega)$ satisfies $I^*(x;\omega) = x^{\eta_{5,BP}+o(1)}$. They use I^* as a proxy for the integer count $\pi_{u,5}(x)$ (their Definition 7.7), counting every integer reaching u under the un-accelerated map $n \mapsto (5n+1)/2$ for n odd, $n/2$ for n even, which they call T_5; they reserve U_5 for the accelerated map this paper calls T_q. That $\pi_{u,5}$ shares $B[5^0]$'s exponent is their unproven heuristic. Separately, collapsing the 2-adic halving chain between consecutive odd values under $\pi_{u,5}$ changes its count by a factor at most $1 + \log_2 x$, which is $x^{o(1)}$, so $N_u(x)$ above shares $\pi_{u,5}(x)$'s exponent η whenever the latter exists. That η exists and does not depend on u is itself their Conjecture 7.2.

Lemma 2.2 (Sibling congruence)**.** *Let $q \geq 3$ be odd, $d = \operatorname{ord}_q(2)$, and v odd with $2^a v \equiv 1 \pmod q$ solvable. Write the children of v, in order of increasing admissible exponent, as $w_{(j)} := w_{a_0+jd} = (2^{a_0+jd}v - 1)/q$ for $j \geq 0$, where $a_0 = a_0(v)$. Then*

$$w_{(j+1)} = 2^d w_{(j)} + \frac{2^d - 1}{q} \qquad \textit{(exact identity in } \mathbb{Z}\textit{)},$$

and consequently $w_{(j+1)} \equiv w_{(j)} + c \pmod q$, with $c := \big((2^d - 1)/q\big) \bmod q$.

Proof. Direct substitution: $2^d w_{(j)} + (2^d - 1)/q = \big(2^{a_0+(j+1)d}v - 2^d + 2^d - 1\big)/q = w_{(j+1)}$. The congruence follows from $2^d \equiv 1 \pmod q$. $\square$

When q is prime, $c = 0$ would force $q^2 \mid 2^d - 1$, the base-2 Wieferich condition; the only known Wieferich primes are 1093 and 3511, so $c \neq 0$ for every q of interest here, and consecutive siblings' residues mod q advance by a fixed nonzero step. At $q = 5$, $d = 4$, $c = 3$. Two facts survive when q is allowed to become a real value: dropping the additive constant, negligible once $w_{(j)}$ is large, leaves consecutive siblings at multiplicative distance $w_{(j+1)} \approx 2^d w_{(j)}$, carrying no factor of q; and every parent-to-child step itself carries the denominator, $w_a \approx 2^a v/q$, also dropping the additive constant. The residue step c stays integer, mod the original q. §5 uses both facts to build constructions with the arithmetic tree's sibling spacing but a tunable exponent.

3 Method

We enumerate T_5's exact reverse tree from 300 roots. They are odd integers coprime to 5 and outside every cycle of T_5 (a cycle member is reachable only from inside its own cycle, so excluding them needs no visited set), sampled uniformly below 10^4 and no smaller than 101, well below the fixed measurement window, 10^5 through 10^8. A single depth-first pass tracks the running path-maximum at every node, giving simultaneous counts at truncation buffers 10^9 through 10^{13} directly, without re-running the enumeration once per buffer; this agrees, count for count, with an independent buffer-by-buffer implementation on five of the 300 roots. The pooled mean slope across the 300 roots rises with truncation. Its increments shrink geometrically:

$$0.60049\ (10^9) \to 0.62387\ (10^{10}) \to 0.63261\ (10^{11}) \to 0.63650\ (10^{12}) \to 0.63801\ (10^{13}),$$

by a factor of 0.37 to 0.44 per decade, the signature of an extrapolable truncation bias.

4 Result: a fixed-window estimator

Aitken's Δ^2 extrapolation of this sequence to infinite truncation gives

$$\hat{\eta}_5 = 0.639, \qquad 95\%\ \text{bootstrap interval, fixed window} = [0.633,\ 0.645]. \tag{1}$$

The bootstrap interval measures variation between sampled roots after the truncation extrapolation; it does not include the remaining bias from the fixed measurement window, and so cannot exclude either asymptotic prediction. That remaining bias is visible directly in a diagnostic panel: on 40 of the 300 roots, at the deepest buffer 10^{13}, the per-decade slope across $10^4 \to 10^5$ through $10^7 \to 10^8$ is still rising at the last decade tested, $0.6021 \to 0.6296 \to 0.6432 \to 0.6460$, with no plateau.

Empirical Result 4.1 (Finite-window reading of the counting exponent)**.** *The fixed-window estimator of the $5x+1$ reverse-tree counting exponent, computed by exact enumeration with Aitken-Δ^2-extrapolated truncation correction, is 0.639 (bootstrap interval $[0.633, 0.645]$). The sequence of local slopes rises monotonically across the available decades, still increasing at the last one tested; §5 extrapolates a related per-decade statistic at far greater truncation depth. On its own, this experiment does not provide a calibrated confidence interval for the asymptotic exponent and does not exclude the Volkov prediction.*

5 A calibrated comparison

The estimator behind Empirical Result 4.1 is biased. An *unconstrained construction* shares the arithmetic tree's offspring law but draws an independent residue at every node; built so its annealed pressure root is the Kontorovich–Lagarias value 0.650919 [2], and read with the same window, truncation, and extrapolation as §4, it gives 0.6131. Reading that gap as bias presumes the construction's realized exponent equals the pressure root it was built to, a presumption examined in §6; taken so, it is an under-read of 0.038, larger than the gap $\Delta = 0.027$ separating the two disputed values.

The bias falls with depth, so comparing readings at a common depth removes that dependence; the residual dependence on how much a process fluctuates is absorbed into the band width read below. That comparison uses a different statistic: the single-decade slope $\log_{10}(N(10^{k+1})/N(10^k))$, Aitken-extrapolated over the truncation buffers above 10^{k+1} and read at a checkpoint decade $10^k \to 10^{k+1}$ deep enough for the bias to have mostly decayed. *Congruence-matched constructions* share the arithmetic tree's offspring law and, via Lemma 2.2, its sibling spacing, with a drawn residue at each parent: an integer recursion, built to exponent 0.650919 only, since it keeps q integer; and its *real-valued relaxation*, built to either target. Only the relaxation's exponent is tunable. Replacing the integer denominator q in the parent-to-child step $w_a \approx 2^a v/q$ by a tunable real value q_{val} moves the relaxation's annealed pressure root to any value α_- (the smaller root) solving $q_{\mathrm{val}}^{\alpha} = q(2^{\alpha} - 1)$. The integer q itself stays put, so the arithmetic tree's own admissibility rule and sterility probability carry over unchanged. Three congruence-matched constructions result: the integer recursion and its relaxation, both built to 0.650919, and that same relaxation retuned to 0.678. The unconstrained construction of the paragraph above, sharing only the offspring law and drawing an independent residue at every node, bounds how much of the calibration depends on the sibling congruence. Every construction below runs on one common set of 300 roots, sampled by the same rule as §3, at checkpoints 10^4 through 10^{10} and truncation buffers 10^9 through 10^{15}, and the arithmetic tree's reading is checked against them. Each root's own drawn residue is constrained fertile, matching the arithmetic tree's roots (sampled coprime to q). No construction starts from a sterile root.

Empirical Result 5.1 (Calibrated separation from the Volkov exponent)**.** *At the checkpoint decade* $10^9 \to 10^{10}$*, the two constructions built to exponent* 0.650919 *that share the sibling congruence read* 0.65122 *(integer recursion) and* 0.64981 *(its real-valued relaxation), a band* 0.00141 *wide. The unconstrained construction built to the same exponent reads* 0.64751*,* 0.0034 *below target; including it widens the band to* 0.0037*. The arithmetic tree reads* 0.64926*, below both constructions built to* 0.650919 *that share the sibling congruence, and inside the wider band set by the unconstrained one. A construction built to exponent* 0.678 *reads* 0.67748*,* 0.0282 *away from the arithmetic reading, more than seven widths of that wider band. The same comparison at the shallower decade* $10^7 \to 10^8$*, on the same checkpoint grid, gives an arithmetic reading of* 0.64622 *against* 0.67200 *for the exponent-*0.678 *construction, a gap of* 0.0258*: the separation grows with depth. This measurement targets the exponent value* 0.678*; Volkov's branching model itself uses a different encoding of the iterates and is not implemented here.*

Pushed on the same 300 roots to checkpoints 10^{12} and buffers 10^{17}, past the depth where a single unlucky realization among the stochastic constructions dominates the wall time of a matched batch, the arithmetic tree's slope alone approaches 0.650919: 0.6465 at $10^7 \to 10^8$, 0.6487 at $10^8 \to 10^9$, 0.6490 at $10^9 \to 10^{10}$, then 0.6506 and 0.6505 at $10^{10} \to 10^{11}$ and $10^{11} \to 10^{12}$, the last two within 0.0005 of the prediction. This deeper grid, with truncation buffers extended to 10^{17}, reads 0.6490 at $10^9 \to 10^{10}$ on the same roots and checkpoints as Empirical Result 5.1: close to, but not identical to, its 0.64926. The bootstrap bands on these five readings cover root resampling only; a separate truncation term is at most 0.002, falling to at most 0.0004 from $10^8 \to 10^9$ onward.

6 Discussion

Two systematics touch Empirical Result 5.1. On the window used for the opening measurement above, the congruence-matched integer recursion and its real-valued relaxation, both built to exponent 0.650919, differ by 0.0035 ± 0.0023 over six seeds (one standard error). That is indistinguishable from zero, and comparable in size to the 0.0037 wider band width itself if it is not. Whether it transfers unchanged to the single-decade slope of the calibration comparison is not tested here.

Separately, the same buffer-headroom test used on the deep run above, applied to the b15 grid, has no slack left at the calibration decade itself. It returns 0.0000 identically for every construction there, uninformative about truncation error. The test has slack at four shallower decades; only $10^7 \to 10^8$, the shallow decade Empirical Result 5.1 reads, is used elsewhere in this comparison, and there it moves each reading by at most 0.0017. The two decades below that move by more, up to 0.0189; the remaining decade, $10^8 \to 10^9$, moves by at most 0.0009. None of their readings appear in this comparison. The trend at $10^7 \to 10^8$ is consistent with the deep run's own fall to at most 0.0004 by $10^8 \to 10^9$ and below that at greater depth, well below the 0.0282 separation. At the calibration depth itself, the relaxation built to exponent 0.678 under-reads its own target by 0.00052 and the relaxation built to 0.650919 by 0.00111.

The 0.038 bias of the opening measurement in §5 and the residual bias at the calibration depth, under 0.004 for every construction there, come from related log-slope estimators of the same family: one over the wider window at shallow depth, one over a single decade at much greater depth. Both the added depth and the narrower window contribute to the drop.

Collapsing the chain of doublings in Kontorovich and Lagarias's $B[5^0]$ gives each exponent $a \geq 1$ an independent chance $1/5$ of a branch at $2^a v/5$: this accelerated form of their model shares the unconstrained construction's mean offspring intensity, hence the same annealed pressure function, and by the companion paper's first-moment argument the same upper bound on the exponent [2]. Equality with that pressure root, which the bias language above presumes, is standard for the unconstrained construction, whose reproduction is i.i.d. across nodes; for the congruence-matched constructions, whose reproduction is correlated by the sibling recursion, it is this paper's working hypothesis. The unconstrained construction and the collapsed $B[5^0]$ differ in their joint structure: the former draws one residue class per node, while the latter assigns each exponent independently.

On the b13 grid of §5's opening measurement, buffers 10^9 through 10^{13}, the unconstrained construction's bias is 0.038. The arithmetic tree's window reading there is 0.63824, a bias of 0.013 presuming the disputed exponent is 0.650919. This figure comes from an independently drawn sample of 300 roots, by the same rule as §4's 0.639; the two values are near but distinct. The deep arithmetic-only run sits systematically at or just below 0.650919, a pattern the data here cannot yet distinguish from a slowly converging sequence.

Data Availability Statement

Code and data reproducing every claim in this paper, with a README per subfolder naming the result it supports, are at `https://github.com/faculdade/collatz-kl-volkov`.

Acknowledgments

AI assistance (Claude, Anthropic) was used for textual review and translation.

References

[1] A. V. Kontorovich and J. C. Lagarias, *Stochastic models for the* $3x + 1$ *and* $5x + 1$ *problems*, arXiv:0910.1944 (2009); published in: *The Ultimate Challenge: the* $3x + 1$ *Problem* (J. C. Lagarias, ed.), American Mathematical Society, 2010. Theorem 8.10 and the Remark following

it (§8.10, including the quotation on Volkov's data), Definition 7.7, and Conjecture 7.2 with the quotation following it (§7) are all cited from the arXiv version.

[2] R. A. Tavares, *A closed-form pressure equation for the accelerated* $qx + 1$ *branching process*, companion paper, in preparation.

[3] S. Volkov, *A probabilistic model for the* $5x + 1$ *problem and related maps*, Stochastic Processes and their Applications **116**, no. 4 (2006), 662–674.